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Quarterly Iteration Cycle Runbook

Operational runbook for the ongoing quarterly swarm iteration cycle. Covers the 6-agent re-execution cadence (CF-01, GL-02, AP-03, NP-04, SA-05, TS-10), the case-file harvest template for replacing synthesized cases with real funded-loan data, the scoring recalibration protocol, and the A/B test learning integration loop.

DOCUMENT TYPE	Operational Runbook
AUDIENCE	Marketing-Ops (1 FTE allocated)
CLASSIFICATION	Internal · Ongoing · Quarterly Cadence
CADENCE	Quarterly (Q2: April, Q3: July, Q4: October, Q1: January)
DURATION	2 weeks per quarter (10 business days)
PRIMARY OUTPUT	Updated persona library + recalibrated scoring weights + refreshed case files
CRITICAL RULE	Never deploy unvalidated scoring changes mid-quarter — always at quarter boundary

SECTION 1

Why a Quarterly Iteration Cycle

The DSCR borrower-intelligence swarm is not a one-time build. It is a living system that must be refreshed quarterly to remain accurate. Three forces drive the need for iteration: first, the evidence base degrades over time as lender programs change, new lenders enter the market, and regulatory shifts alter the fundability surface; second, the persona library drifts as funded-loan data reveals which personas actually convert versus which were hypothesized to convert; third, the scoring engine miscalibrates as the ad platforms learn and as market conditions shift. Without quarterly iteration, the swarm's cost per funded loan will degrade 15-25 percent per quarter, and within 4 quarters the swarm will be operating on stale assumptions that no longer reflect market reality. The quarterly iteration cycle is the swarm's learning loop: every funded loan and every declined file feeds back into the evidence base, sharpening the persona library, the scoring weights, and the targeting precision.

Iteration Cycle Cadence

Quarter	Iteration Window	Pre-Iteration Data Cutoff	Post-Iteration Deployment	Marketing-Ops FTE Allocation
Q2 (Apr-Jun)	April 1-12	March 31 funded-loan + decline data	April 15 (Monday after iteration)	1.0 FTE for 2 weeks
Q3 (Jul-Sep)	July 1-12	June 30 data	July 15	1.0 FTE for 2 weeks
Q4 (Oct-Dec)	October 1-12	September 30 data	October 15	1.0 FTE for 2 weeks
Q1 (Jan-Mar)	January 1-12	December 31 data	January 15	1.0 FTE for 2 weeks

CRITICAL RULE

NEVER deploy scoring-engine changes mid-quarter. Scoring changes always deploy at the quarter boundary (April 15, July 15, October 15, January 15). Mid-quarter changes corrupt A/B test data and make it impossible to attribute funded-loan outcomes to specific scoring configurations. If a critical scoring bug is discovered mid-quarter, deploy a hotfix ONLY if the bug is causing leads to be routed incorrectly (e.g., Tier A leads routed to Tier C queue). Otherwise, queue the fix for the next quarter boundary.

SECTION 2

Six-Agent Re-Execution Sequence

Each quarterly iteration re-executes 6 of the 10 swarm agents (CF-01, GL-02, AP-03, NP-04, SA-05, TS-10) on the updated evidence base. The 4 agents not re-executed (EG-06, GS-07, FF-08, AC-09) are refreshed annually (Q2 each year) unless a triggering event requires mid-year refresh. The re-execution follows the same dependency graph as the original build: evidence first, then patterns, then personas, then scoring.

Day	Agent	Activity	Input	Output	Duration
Day 1-2	CF-01 Case File Harvester	Re-harvest case files: replace synthesized cases with real funded-loan data from prior quarter + harvest new public case studies	CRM funded-loan data + public lender case studies	Updated case file set (target: 50+ real cases, <25% synthesized)	2 days
Day 3	GL-02 Guideline Normalizer	Re-normalize lender guidelines: capture program changes, new entrants, withdrawn programs	Lender websites + industry publications	Updated guideline matrix + new compensating-factor patterns	1 day
Day 4-5	AP-03 Approval Pattern Miner	Re-mine approval patterns from updated case files + guidelines	CF-01 output + GL-02 output	Updated approval clusters + FDI recalculations	2 days
Day 5	NP-04 Negative Pattern Miner	Re-mine decline patterns: capture new decline drivers, retire obsolete ones	CF-01 output + GL-02 output	Updated decline clusters + HEX rule audit	1 day (parallel with AP-03)
Day 6-7	SA-05 Sponsor Archetype Synthesizer	Update persona library: retire low-FDI personas, add newly-discovered personas, refresh FDI scores	AP-03 output + NP-04 output	Updated persona library (target: 12-15 personas)	2 days
Day 8-9	TS-10 Targeting & Scoring Generator	Recalibrate scoring engine: adjust component weights, update modifier triggers, refresh tier-routing thresholds	SA-05 output + funded-loan cohort data from prior quarter	Updated scoring engine + recalibration report	2 days
Day 10	All	Cross-agent validation + deployment prep + A/B test calendar update	All outputs	Iteration deployment package	1 day

Annual Refresh (Q2 Each Year)

In addition to the 6-agent quarterly refresh, Q2 each year includes a full 10-agent refresh that re-executes EG-06 (edge-case gold miner), GS-07 (geo-segment correlator), FF-08 (funnel friction mapper), and AC-09 (ad hook & copy reframer). The annual refresh is a 3-week exercise (April 1-21) and produces a complete swarm refresh. The annual refresh is required because edge-case personas, geo-targeting, intake friction points, and ad creative all evolve more slowly than the core evidence base but still require periodic refresh to remain accurate.

SECTION 3

Case File Harvest Template (CF-01 Quarterly)

The CF-01 quarterly re-harvest replaces synthesized cases with real funded-loan data. The CRM team extracts funded-loan data from the prior quarter and structures it into the CF-01 case file YAML schema. The target is 50+ real cases per quarter, reducing the synthesized-case ratio below 25 percent by end of Q2.

Funded-Loan Data Extraction Query (CRM)

```
-- CRM SQL query: extract funded loans from prior quarter
-- Run on April 1 (Q2 iteration), July 1 (Q3), October 1 (Q4), January 1 (Q1)

SELECT
l.loan_id,
l.funded_date,
l.loan_amount,
l.interest_rate,
l.lender_name,
l.property_type,
l.property_state,
l.property_msa,
l.occupancy_type,
l.dscr_at_funding,
l.fico_at_application,
l.ltv_at_funding,
l.reserves_months,
l.entity_structure,
l.borrower_experience_doors,
l.borrower_employment_type,
l.persona_match,
l.tier_at_application,
l.time_to_close_days,
l.specialty_lender_flag,
l.decline_letter_audit_flag,
l.notes
FROM loans l
WHERE l.funded_date >= DATEADD(quarter, -1, CURRENT_DATE)
AND l.funded_date < CURRENT_DATE
AND l.product_type = 'DSCR'
ORDER BY l.funded_date DESC;
```

Case File YAML Template (CF-01 Schema)

Each funded loan is structured into the CF-01 YAML schema. The CRM team generates this YAML automatically from the SQL query output, then marketing-ops reviews for accuracy.

```
case_id: CF-Q2-2026-001 # Quarter + Year + sequence
source_type: internal_funded_loan # NOT "synthesized"
synthesized: false
borrower_profile:
```

```
experience_level: 6_20_doors # from borrower_experience_doors
employment: self_employed # from borrower_employment_type
entity: LLC # from entity_structure
state: TX # from property_state
property:
type: SFR # from property_type
occupancy: long_term_rental # from occupancy_type
market: Houston # from property_msa
purchase_price_or_value: <USD from loan_amount / LTV>
estimated_rent: <USD from rent schedule at funding>
loan_request:
purpose: purchase # or refi_rate_term / refi_cashout
loan_amount: <USD>
LTV: <percent from ltv_at_funding>
underwriting_inputs:
DSCR: <ratio from dscr_at_funding>
FICO: <score from fico_at_application>
reserves_months: <number from reserves_months>
dti_if_disclosed: n/a # DSCR loans do not use DTI
outcome: approved # funded loans are always approved
approval_conditions: [list from notes]
source_quote: "<verbatim from notes if borrower quote available>"
notes: <analyst commentary — what made this file fundable>
funded_loan_economics:
time_to_close_days: <from time_to_close_days>
specialty_lender_flag: <from specialty_lender_flag>
decline_letter_audit_flag: <from decline_letter_audit_flag>
```

Decline-File Harvest (Parallel)

In addition to funded loans, the swarm needs decline data to refresh NP-04. CRM team extracts decline files from the prior quarter using the query below. Decline files are particularly valuable for identifying new decline drivers and false-positive risks.

```
-- CRM SQL query: extract declined files from prior quarter
SELECT
l.lead_id,
l.created_date,
l.decline_date,
l.decline_reason_primary,
l.decline_reason_secondary,
l.persona_match,
l.tier_at_application,
l.property_type,
l.property_state,
l.dscr_estimate,
l.fico_band,
l.ltv_estimate,
l.reserves_months_estimate,
l.entity_structure,
l.borrower_experience_doors,
l.decline_letter_audit_outcome,
l.specialty_lender_referral_flag,
```

```
l.notes
FROM leads l
WHERE l.status = 'declined'
AND l.decline_date >= DATEADD(quarter, -1, CURRENT_DATE)
AND l.decline_date < CURRENT_DATE
AND l.product_type = 'DSCR'
ORDER BY l.decline_date DESC;
```

SECTION 4

Scoring Recalibration Protocol (TS-10 Quarterly)

The TS-10 scoring engine is recalibrated quarterly based on observed funded-loan outcomes. The recalibration protocol is data-driven and conservative: weights adjust only when funded-loan cohort data shows the current weights are mispredicting outcomes. The protocol prevents over-fitting to small sample sizes and prevents reactionary changes based on short-term noise.

Recalibration Triggers

Trigger	Condition	Action	Sample Size Requirement
T-1: Tier A conversion below target	Tier A leads convert to funded loans at <60% over the quarter	Increase SC-001 (DSCR) and SC-005 (Property Type) weights by 2-3pts each; decrease SC-007 (Experience) by 2-3pts	Minimum 30 Tier A leads in cohort
T-2: Tier C conversion above target	Tier C leads convert to funded loans at >25% over the quarter	Lower Tier C threshold from 40 to 35 (more leads route to specialty); add 2-3pts to SC-008 (Edge-Case Bonus)	Minimum 20 Tier C leads in cohort
T-3: Tier D false-positive	Tier D leads (auto-declined at intake) fund through manual override at >10%	Audit HEX rules for over-filtering; add remediation paths for affected HEX rules	Minimum 50 Tier D leads in cohort
T-4: Persona FDI drift	Any persona's observed FDI (funded-loan conversion) differs from predicted FDI by >2 points	Recalculate persona FDI using observed data; retire personas with observed FDI <5.0; add new personas if edge-case conversion data supports	Minimum 10 funded loans per persona
T-5: New decline driver	NP-04 identifies a new decline driver not in the current HEX/SWR rule set	Add new HEX or SWR rule; update FF-08 intake if needed	Minimum 5 cases with the new decline driver
T-6: Lender program change	GL-02 identifies a lender program change (new LTV cap, new FICO floor, withdrawn program)	Update TS-10 scoring logic for affected lenders; update SA-05 persona lender lists	Confirmed program change from lender

Recalibration Workflow

The recalibration workflow is a 4-step process that runs on Day 8-9 of the iteration cycle:

- **Step 1 (Day 8 AM): Cohort data extraction** — Extract funded-loan + decline cohorts from prior quarter. Calculate observed conversion rates by tier, by persona, by geo, by channel.
- **Step 2 (Day 8 PM): Trigger evaluation** — Evaluate each of the 6 triggers (T-1 through T-6) against the cohort data. Document which triggers fire and the proposed recalibration action.
- **Step 3 (Day 9 AM): Conservative adjustment** — Apply recalibration actions conservatively. No weight changes greater than 3pts per component per quarter. No threshold changes greater than 5pts per quarter. Document the before/after for each change.
- **Step 4 (Day 9 PM): A/B test design** — Design an A/B test for each recalibration change. The A/B test runs for 4 weeks (rest of April for Q2 iteration) with the old scoring as control and new scoring as variant. If the variant does not outperform control by week 4, revert to old scoring for the next quarter.

Recalibration Report Template

```
# TS-10 Quarterly Recalibration Report — Q2 2026

## Cohort Data
- Funded loans (Q1 2026): 18
- Declined files (Q1 2026): 142
- Total leads (Q1 2026): 387
- Tier distribution: A=42 (11%), B=98 (25%), C=72 (19%), D=175 (45%)

## Trigger Evaluation
- T-1 (Tier A conversion): Tier A converted 23/42 = 54.8%. BELOW 60% target. TRIGGER
FIRES.
- Proposed action: Increase SC-001 (DSCR) weight from 25 to 27; increase SC-005 (Property
Type) from 10 to 12; decrease SC-007 (Experience) from 5 to 3.
- T-2 (Tier C conversion): Tier C converted 11/72 = 15.3%. Below 25% target. Does NOT
fire.
- T-3 (Tier D false-positive): Tier D manual-override funding 4/175 = 2.3%. Below 10%
threshold. Does NOT fire.
- T-4 (Persona FDI drift): SA-007 STR Operator observed FDI = 6.2 (predicted 7.6, drift
-1.4). Below 2pt threshold. Does NOT fire but monitor next quarter.
- T-5 (New decline driver): NP-04 identified "insurance-unavailable in FL coastal" as new
decline driver. TRIGGER FIRES.
- Proposed action: Add SWR-015 "FL coastal insurance check" to TS-10 modifiers (-5 score).
- T-6 (Lender program change): Newfi withdrew 85% LTV program; max now 80%. TRIGGER FIRES.
- Proposed action: Update SC-003 scoring logic for Newfi referrals; cap at 80% LTV.

## Recalibration Actions Applied (Conservative)
1. SC-001 weight: 25 -> 27 (+2, within 3pt limit)
2. SC-005 weight: 10 -> 12 (+2, within 3pt limit)
3. SC-007 weight: 5 -> 3 (-2, within 3pt limit)
4. SWR-015 added: FL coastal insurance check, -5 score modifier
5. SC-003 logic updated: Newfi max LTV capped at 80% (was 85% for premium files)

## A/B Test Design (Q2 2026, Weeks 2-6)
- Test ID: T-Q2-001
- Variant: New scoring (with recalibration applied)
- Control: Old scoring (Q1 2026 configuration)
- Primary metric: Tier A conversion to funded loan
- Sample size: Minimum 30 Tier A leads per arm
- Duration: 4 weeks (April 15 - May 13)
- Decision rule: If variant outperforms control by >=5pts in Tier A conversion, deploy
variant for Q3. If <5pts, revert to control for Q3.
```

SECTION 5

A/B Test Learning Integration Loop

The A/B test calendar from the 90-day rollout (TS-10 Part 3) produces learning that must be integrated into the next quarter's swarm configuration. The integration loop runs on Day 10 of the iteration cycle and consolidates A/B test winners into the persona library, scoring weights, and creative rotation.

Learning Integration Workflow

- **Step 1: Consolidate A/B test winners** — Review all A/B tests from the prior quarter. Document winners (statistically significant at 95% confidence, minimum 100 conversions per variant). Document inconclusive tests (extend or drop).
- **Step 2: Update creative rotation** — Replace losing hook variants with winning hook variants in the Meta and Google ad sets. Add 1-2 new variants per persona for the next quarter's test calendar.
- **Step 3: Update landing page variants** — Deploy winning landing-page variants as the new default. Retire losing variants. Add 1-2 new variants per persona for the next quarter's test calendar.
- **Step 4: Update scoring weights (if test supports)** — If an A/B test specifically tested scoring weights (e.g., T-Q2-001 from the recalibration report), deploy the winning configuration. If the test was inconclusive, revert to control.
- **Step 5: Update persona FDI scores** — If A/B test data shows a persona's creative is significantly underperforming or outperforming, adjust the persona's marketing_reachability score (one of the 7 FDI dimensions). Do NOT adjust other FDI dimensions based on creative performance.
- **Step 6: Document learning** — Add a learning log entry to the swarm knowledge base. Each entry: test ID, hypothesis, result, action taken, follow-up test (if any).

A/B Test Learning Log Template

```
# Swarm A/B Test Learning Log

## T-001 (M1, Meta, SA-001, Hook category test)
- Hypothesis: PI hooks outperform PA hooks on Meta for SA-001
- Variants: PI (H1, H2) vs PA (H3, H4) vs PS (H5, H6)
- Result: PA hooks won with 32% lower cost per Tier A-or-B lead (stat sig 95%, 142
conversions per variant)
- Action: Replace PI hooks with PA hooks as default for SA-001 Meta campaigns in Q2
- Follow-up: Test new PA hook variants (H7, H8) against winning H3/H4 in Q2

## T-002 (M1, Meta, SA-002, Hook category test)
- Hypothesis: PS hooks outperform PI hooks on Meta for SA-002
- Variants: PI vs PA vs PS
- Result: Inconclusive at 4 weeks. Extended to 6 weeks. Still inconclusive. Dropped.
- Action: No change. All 6 variants continue rotation in Q2.
- Follow-up: Test LinkedIn-only PS hook (H7) vs Meta PS hook (H5) in Q2

## T-007 (M3, Landing page, SA-001, Hook A vs B vs C)
- Hypothesis: V2 hook C outperforms A and B on landing-page form-start rate
- Variants: A (PI), B (PA), C (PS)
- Result: Hook C won with 18% higher form-start rate (stat sig 95%, 87 form-starts per
variant)
- Action: Deploy hook C as default SA-001 landing page in Q2
- Follow-up: Test hook C above-fold vs below-fold in Q2

[... continue for all 9 tests ...]
```

SECTION 6

Iteration Quality Gates

Each quarterly iteration must pass 5 quality gates before deployment. Gate failures delay deployment by one week per failed gate, with a maximum 2-week delay before the iteration is rescheduled for the next quarter.

Gate	Criterion	Owner	Failure Action
QG-1: Real-case ratio	CF-01 output has >=50% real cases (not synthesized)	Marketing Ops	Extend CF-01 harvest by 1 week; if still <50%, deploy with reduced confidence and flag for next quarter
QG-2: Persona FDI validation	SA-05 output shows all personas with FDI >=5.0; no persona retired without documented evidence	Marketing Ops + LO Ops	If persona retired, document evidence; if FDI<5.0, retire and re-allocate budget
QG-3: Scoring weight delta	TS-10 weight changes are within +/-3pts per component; threshold changes within +/-5pts	Marketing Ops + Data	If exceeds limits, reduce changes to within limits; document the conservative reduction
QG-4: A/B test statistical significance	All A/B test winners reached 95% confidence + 100 conversions per variant	Marketing Ops	Inconclusive tests dropped; winners only deployed
QG-5: Compliance re-review	Any new persona, any new HEX rule, any scoring weight change >2pts requires compliance review	Compliance	If compliance review not complete by Day 10, delay deployment 1 week

Iteration Deployment Checklist

Run this checklist on Day 10 before deploying the iteration:

- ☒ QG-1 through QG-5 all passed (or failure actions documented)
- ☒ CF-01 case file set updated and committed to swarm knowledge base
- ☒ GL-02 lender guideline matrix updated
- ☒ AP-03 approval clusters refreshed with FDI recalculations
- ☒ NP-04 decline clusters refreshed with new HEX/SWR rules
- ☒ SA-05 persona library updated (new personas added, retired personas documented)
- ☒ TS-10 scoring engine recalibrated with documented before/after
- ☒ A/B test learning log updated with prior-quarter winners
- ☒ A/B test calendar loaded for new quarter (9 new tests minimum)

- ☒ CRM scoring engine configuration updated and unit tested
- ☒ Meta ad sets updated with winning creative variants
- ☒ Google ad groups updated with winning keywords and landing pages
- ☒ KPI dashboard refreshed with prior-quarter actuals vs targets
- ☒ Iteration deployment package reviewed and approved by marketing-ops lead
- ☒ Compliance re-review complete (if triggered by QG-5)

SECTION 7

Year-1 Swarm Health Targets

The swarm's health is measured against year-1 targets. These targets assume the 90-day rollout hits its KPIs and the quarterly iteration cycle is executed faithfully. Missing year-1 targets triggers a swarm health review and potential architectural changes.

Metric	Q1 2026 (M1-M3)	Q2 2026	Q3 2026	Q4 2026	Year-1 Total
Funded loans	5	12	18	24	59
Cost per funded loan	<$8,500	<$7,500	<$6,500	<$6,000	<$7,000 blended
Tier A-or-B leads	390	520	650	780	2,340
Cost per Tier A-or-B lead	$250	$230	$210	$200	$220 blended
Real-case ratio (CF-01)	25%	50%	65%	75%	75% by year-end
Persona library size	12	13	14	15	15 by year-end
Specialty-lender referral rate	20%	25%	28%	30%	30% by year-end
A/B tests run	9	12	12	12	45
A/B test winners deployed	6	8	8	8	30
Quarterly iteration completed	—	Yes	Yes	Yes	3 iterations

Year-1 Swarm Health Review (January 2027)

In January 2027, marketing-ops conducts a full year-1 swarm health review. The review covers: (1) did the swarm hit year-1 targets, (2) which personas over-performed vs under-performed, (3) which channels delivered the best cost per funded loan, (4) which specialty lenders delivered the most funded loans, (5) what architectural changes are needed for year-2 (e.g., new agents, new data sources, new channels). The year-1 review is the input to the year-2 swarm roadmap, which is presented to leadership in February 2027.

> **ITERATION CYCLE IS THE SWARM'S COMPOUNDING ADVANTAGE**

> The quarterly iteration cycle is what makes the swarm a strategic asset rather than a one-time marketing campaign. Each quarter, the swarm gets sharper: the persona library reflects real funded-loan data, the scoring engine reflects real conversion outcomes, the creative reflects real A/B test winners. Competitors who run static campaigns will fall behind within 2-3 quarters. The compounding advantage is the swarm's primary defensible value. Protect the iteration cycle cadence — do not let quarterly pressure push the iteration to "next quarter."